

Quantum Fault Tolerance Meets Toom's Contours

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1 Hadamard and S Gates

Claim 1.1. For any generator subset S' of size d' , denote by $x_{S'}$ and $z_{S'}$ the following assignments:

$$x_{S'} := \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} \quad z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, S'}$$

Then $x_{S'}$ and $z_{S'}$ are binary representations of anticommuting logical codewords of the (d, d') -Toric code. In particular, $x_{S'} \in \mathcal{C}_X / \mathcal{C}_Z^\perp$, $z_{S'} \in \mathcal{C}_Z / \mathcal{C}_X^\perp$, and $x_{S'} z_{S'} = 1$. In addition, let $X_{S'}$ and $Z_{S'}$ be the Pauli matrices obtained by multiplying each non-trivial coordinate of them by X or Z , respectively.

Proof. By definition $x_{S'}$ and $z_{S'}$ are binary assignments on the d' -dimensional faces. We start by showing that $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$:

$$\begin{aligned} \partial^- \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u}, S'/g} + \theta_{\mathbf{g}\mathbf{u}, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}\mathbf{u}, S'/g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'/g} = 0 \\ \partial^+ \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S'} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u}, S/S' \cup g} + \theta_{\mathbf{g}^{-1}\mathbf{u}, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g}^{-1}\mathbf{u}, S/S' \cup g} \\ &= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S/S' \cup g} = 0 \end{aligned}$$

Thus $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabilizers, it's enough to show they admit no vanishing product.

$$x_{S'} z_{S'} = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, S'} \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, S'} = \theta_{\mathbf{e}, S'} \theta_{\mathbf{e}, S'} = 1$$

□

Definition 1.2 (The Transpose Map). Let d, d' be integers, and let $\mathcal{G}$ be a (d, d') -Toric complex generated by the generator set S . We define the *transpose map* $\phi : \mathcal{F}^{d'} \rightarrow \mathcal{F}^{d-d'}$ by acting on the basis elements as follows:

$$\phi(\theta_{\mathbf{v}, \mathbf{S}'}) \rightarrow \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'}$$

Notice that for $d' = d - d'$, the transpose map is a permutation over the bits.

Definition 1.3 (Hadamard-Type Gate H_ϕ). For $d' = d - d'$, we define the logical Hadamard-type gate $H_\phi : L \rightarrow L$ as follows, for any $S' \subset S$, of size d' :

$$H_\phi(X_{S'}) \rightarrow Z_{S/S'}$$

$$H_\phi(Z_{S'}) \rightarrow X_{S/S'}$$

Claim 1.4. The Hadamard-type gate H_ϕ has a fold-transversal implementation by, first applying H^n over all the physical qubits, and then permuting them by the transpose map ϕ , namely $\hat{H}_\phi = \phi \circ H^n$.

Proof. Clearly, for any subset of generators $S' \subset S$, the map ϕ sends $x_{S'}$ to $z_{S/S'}$:

$$\phi(x_{S'}) = \phi\left(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}'}\right) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}/\mathbf{S}'} = z_{S/S'}$$

Moreover, ϕ sends binary representations of Z -stabilizers to binary representations of X -stabilizers, and vice versa. To see this, consider a $d' + 1$ dimensional face, and consider the image of the stabilizer it defines.

$$\begin{aligned} \phi(\partial^- \theta_{\mathbf{v}, \mathbf{S}'}) &= \phi\left(\sum_{g \in S'} \theta_{\mathbf{v}, \mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{g}\mathbf{v}, \mathbf{S}'/\mathbf{g}}\right) \\ &= \sum_{g \in S'} \theta_{\mathbf{v}^{-1}, (\mathbf{S}/\mathbf{S}') \cup \mathbf{g}} + \theta_{\mathbf{g}^{-1}\mathbf{v}^{-1}, (\mathbf{S}/\mathbf{S}') \cup \mathbf{g}} \\ &= \partial^+ \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'} \end{aligned}$$

And since $\phi \circ \phi = I$, we also have that $\phi(\partial^+ \theta_{\mathbf{v}^{-1}, \mathbf{S}/\mathbf{S}'})$. Thus, $H^{\otimes n} \circ \phi$ sends Z -stabilizers to X -stabilizers and vice versa, thereby preserving the code space. Combining this with the above, $H^{\otimes n} \circ \phi$ realizes the logical form H_ϕ in the $(d, d/2)$ -Toric code. $\square$

Corollary 1.5. For the $(d, d/2)$ -Toric code, consider the gate $CZ_{S' \rightarrow S/S'} : L^{\otimes 2} \rightarrow L^{\otimes 2}$, defined to act on two encoded registers by applying $Z_{S/S'}$ on the second register, controlled by whether the logical $X_{S'}$ is turned on in the first register. Then $CZ_{S' \rightarrow S/S'}$ has a fold-transversal realization.

Proof. Consider the following circuit,

1. Apply H^n on the second register.
2. For any bit $\theta_{\mathbf{v}, \mathbf{S}'}$, apply CX from $\theta_{\mathbf{v}, \mathbf{S}'}$ in the first register into $\phi(\theta_{\mathbf{v}, \mathbf{S}'})$ in the second register.

3. Again apply H^n on the second register.

□

Definition 1.6 (The φ Rotation). Consider a permutation from the generator set to itself $\pi: S \rightarrow S$. The permutation π induces a natural isomorphism by substituting generators of the product representation with their corresponding images according to ϕ . We define the φ_π rotation, $\varphi_\pi: \mathcal{F}^{d'} \rightarrow \mathcal{F}^{d'}$, to act on the d' -dimensional faces as follows:

$$\varphi_\pi(\theta_{\mathbf{v}, \mathbf{S}'}) := \theta_{\pi(\mathbf{v}), \pi(\mathbf{S}')}$$

Claim 1.7. Fix a generator subset S' of size d' , and consider a permutation $\pi: S \rightarrow S$, such that $\pi: S' \rightarrow S/S'$.

$$\varphi_\pi(x_S) = \varphi_\pi\left(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u}, \mathbf{S}'}\right) = \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u}, \mathbf{S}/\mathbf{S}'} = x_{S/S'}$$

[COMMENT] Show that the code space is preserved under the application of φ_π .

$$\begin{aligned} \varphi(\partial^- \theta_{\mathbf{v}, \mathbf{J}}) &= \varphi\left(\sum_{g \in J} \theta_{\mathbf{v}, \mathbf{J}/\mathbf{g}} + \theta_{\mathbf{g}\mathbf{v}, \mathbf{J}/\mathbf{g}}\right) \\ &= \sum_{g \in J} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J}/\mathbf{g})} + \theta_{\varphi(\mathbf{g})\varphi(\mathbf{v}), \varphi(\mathbf{J}/\mathbf{g})} \\ &= \sum_{g \in \varphi(J)} \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J})/\mathbf{g}} + \theta_{\mathbf{g}\varphi(\mathbf{v}), \varphi(\mathbf{J})/\mathbf{g}} \\ &= \partial^- \theta_{\varphi(\mathbf{v}), \varphi(\mathbf{J})} \end{aligned}$$

Definition 1.8 (The Nice Encoding). Consider $(d, d/2)$ -Toric code and fix a subset $S' \subset S$, at size d' . We define the *nice encoding* to be the subsystem code, which encodes a single logical qubit as follow:

$$\begin{aligned} |\bar{0}\rangle &= |\mathcal{C}_Z^\perp\rangle \\ |\bar{1}\rangle &= |x_{S'} + x_{S/S'} + \mathcal{C}_Z^\perp\rangle \end{aligned}$$

Claim 1.9. Given a resource state, the nice encoding has a fold-transversal realization of the logical $\mathbf{S}$ gate.

Proof. Consider the resource state defined by:

$$|\mathbf{S}\rangle = \frac{1}{\sqrt{2}} (|\bar{0}\rangle + i|\bar{0} + x_{S'}\rangle)$$

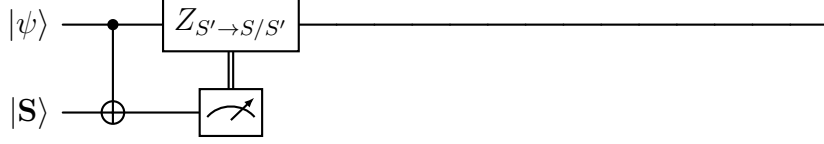


Figure 1: Simulating the **S**-gate, using the $|\mathbf{S}\rangle$ magic state.

$$\begin{aligned}
|\psi\rangle |\mathbf{S}\rangle &= (\alpha |\bar{0}\rangle + \beta |\bar{0} + x_{S'} + x_{S/S'}\rangle) \frac{1}{\sqrt{2}} (|\bar{0}\rangle + i |\bar{0} + x_{S'}\rangle) \\
&\mapsto \frac{1}{\sqrt{2}} (\alpha |\bar{0}\rangle |\bar{0}\rangle + i\alpha |\bar{0}\rangle |\bar{0} + x_{S'}\rangle \\
&\quad + \beta |\bar{0} + x_{S'} + x_{S/S'}\rangle |\bar{0} + x_{S'} + x_{S/S'}\rangle + i\beta |\bar{0} + x_{S'} + x_{S/S'}\rangle |\bar{0} + x_{S/S'}\rangle)
\end{aligned}$$

□

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit). Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, \dots, D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o). Let C be a quantum circuit, at depth d , with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, \dots, \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C , denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t + 1$ that support qubits (faces)

such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$E_o := \{ \{(v, t), (u, t')\} : \text{if} \\ \text{either there exists } i \text{ such } \{v, u\} \in \mathcal{G}_i \text{ and } t' = t, t + 1 \\ \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$). For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o . For each time t , denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial\mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are taken to be the vertices of G_o , namely tuples $(v, t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t , belongs to the $\partial\mathcal{D}_t$. Namely:

$$\{v_t : v_t = (v, t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial\mathcal{D}_t\}$$

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where $A_{\mathcal{H}}$ are the edges in E_o restricted to $V_{\mathcal{H}}$ and $F_{\mathcal{H}}$ are defined to be the . Namely:

$$A_{\mathcal{H}} = \{e = \{u, v\} \in E_o : u, v \in V_{\mathcal{H}} \text{ and } T(v) = T(u) \pm 1\} \\ F_{\mathcal{H}} = \{e = \{u, v\} \in E_o : \text{exists } w \in V_{\mathcal{H}} \text{ such } \{v, w\}, \{u, w\} \in A_{\mathcal{H}}\}$$

By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.

Definition 2.4 (**Size()**-Function). Let S be a subset of vertices, we define the *size* of S to be:

$$\mathbf{Size}(S) := \sum_i^{d+1} \max_{v \in S} L_i(v)$$

Claim 2.5. Let $\mathbf{S} = \{S_1, S_2, \dots, S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned} \mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\ &\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\ &= \mathbf{Size}(R_1) + \mathbf{Size}(R_2) \end{aligned}$$

□

Definition 2.6 (Size Invariant Action). [COMMENT] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z , it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.7. The sweep rule is Size ξ -shrinking.

Proof.

□

[COMMENT] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time = $3t$, in that we apply an invariant action so.